

28th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2024)

The concept and method of determining the relation between data using relational equations with multi-operations composition

Zofia Matusiewicz^{a,*}^a*Department of Cognitive Science and Mathematical Modeling, University of Information Technology and Management, Al. Sucharskiego 2, 35–225 Rzeszów, Poland*

Abstract

Discovering knowledge from data has become one of the most critical problems in computer science in the last decades. Many methods and solutions to this issue have been created. It is not only the collection and analysis of data that is becoming an indispensable part of our lives but also the continuous process of improving detection methods for discovering knowledge from data. In the presented work, we modify the study of the relationship between attributes and specific ones, which are fuzzy relational equations. E. Sanchez, one of the pioneers of work on fuzzy relational equations, started research on using this method to study the relationship between input and output data, indicating it as a tool for analysing medical data. Since the 1970s, these equations have been studied with different types of compositions. The author of this work deals with this subject, examining the assumptions regarding the operations $*$ that can be used in $\max - *$ relations' composition in fuzzy relation equations $A \circ x = b$ to have the solution set. In this work, we use a new way of compositing relations. It enables the use of various types of decision-attribute dependencies. We note that various dependencies may exist between individual data attributes and the decision. Undoubtedly, it is another stage of work on relational equations and provides new opportunities to discover the relationships between input and output data.

© 2024 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the 28th International Conference on Knowledge Based and Intelligent information and Engineering Systems

Keywords: fuzzy relational equation, fuzzy systems of equations, discovering knowledge from data;

1. Introduction

The starting point of the rough set and analysis using fuzzy relational equations is the assumption that with every object of interest, we associate some information (data, knowledge) [13], and [21], and also [16], [15]. Very often, the concepts and achievements of fuzzy and rough sets are combined. As pointed out in the article [25], fuzzy rough set theory is a powerful tool for dealing with uncertain information. Previously, an attribute reduction method was pro-

^{*} Corresponding author. Tel.: +48-17-866-11-32; fax: +48-17-866-12-22.E-mail address: zmatusiewicz@wsiz.edu.pl

posed based on relational systems of equations [8], [9]. Nowadays, combined fuzzy-rough methods can be observed, for example, in the work [24].

The input data for Data Mining are presented in the decision table, a so-called data set. In it, each object (case) is described by attributes. Each record is an independent object with its disjoint pairs of variables and a decision. The decision is a value, which is a variable dependent on its attributes. In Pawlak's work, we can meet the following example: six patients with three medical features (attributes) and the decision to have flu (see 1). We can present this

Table 1. An example of a decision table.

Patient	Headache	Muscle pain	Temperature	Flu
p1	no	yes	high	yes
p2	yes	no	high	yes
p3	yes	yes	very high	yes
p4	no	yes	normal	no
p5	yes	no	high	no
p6	no	yes	very high	yes

in a more general form (see Fig. 1). The following steps in data analysis vary. Data analysis looks different when we use the rough set of the already mentioned Pawlak or methods resulting from the theory of fuzzy sets. Undoubtedly, these methods can support each other, complement each other and serve as a pre-processing method for each other.

	attribute 1	attribute 2	...	attribute n	
object 1	value (o_1, a_1)	value (o_1, a_2)		value (o_1, a_n)	decision 1
...					...
object m	value (o_m, a_1)	value (o_m, a_2)		value (o_m, a_n)	decision m

Fig. 1. The form of a data set.

Moreover, as Kerre indicated, at the beginning of exploring fuzzy relational equations' subject [10], the following possible application areas of fuzzy relations were considered: information retrieval, medical diagnosis, and expert system inference mechanisms. Of course, nowadays, we have many such applications. However, in this work, we do not consider possible applications, but we try to solve problems connected with different kinds of dependence between attributes and obtained decisions. Nevertheless, further improving this method of examining data and the relationships between them is a significant research problem.

In analysing, where we use relational systems of equations, it is also necessary to translate the values from linguistic form to fuzzy set. Let us recall that the value of a linguistic variable is qualitative data most often referred to in natural language, for example high, medium, low.

The process of finding relationships between attributes (input data) and a decision (output data) is also called Inverse Problem Applications (see [19]). Input data (attributes) and output data (decisions) are linguistic variables that need to be mapped to the interval $[0, 1]$. Therefore, we must determine the membership function for the set μ_A of the fuzzy A . Let us recall that the fuzzy set A in X is the set of ordered pairs:

$$A = \{(x, \mu_A(x)) : x \in X\}, \quad (1)$$

where $\mu_A : X \rightarrow [0, 1]$. As a result of this process, relationship matrices are prepared. So for sets of objects $O = \{o_1, \dots, o_m\}$ and attributes $Q = \{q_1, \dots, q_n\}$ a relation matrix A is created, and for these objects O and decision for them $D = \{d_1, \dots, d_m\}$ is preparing vector b . So, this matrix's elements a_{ij} have degrees of feature j of object i , and

b_i contains degrees of decision for object i , where $i \in \{1, \dots, m\}$. Further, we search a vector x describing the relation between attributes and decisions. This brings us to solving the relational equation $A \circ x = b$. The whole process is presented in Fig. 2 (see [19]). Figure 1 illustrates the process of finding relationships between input data and output

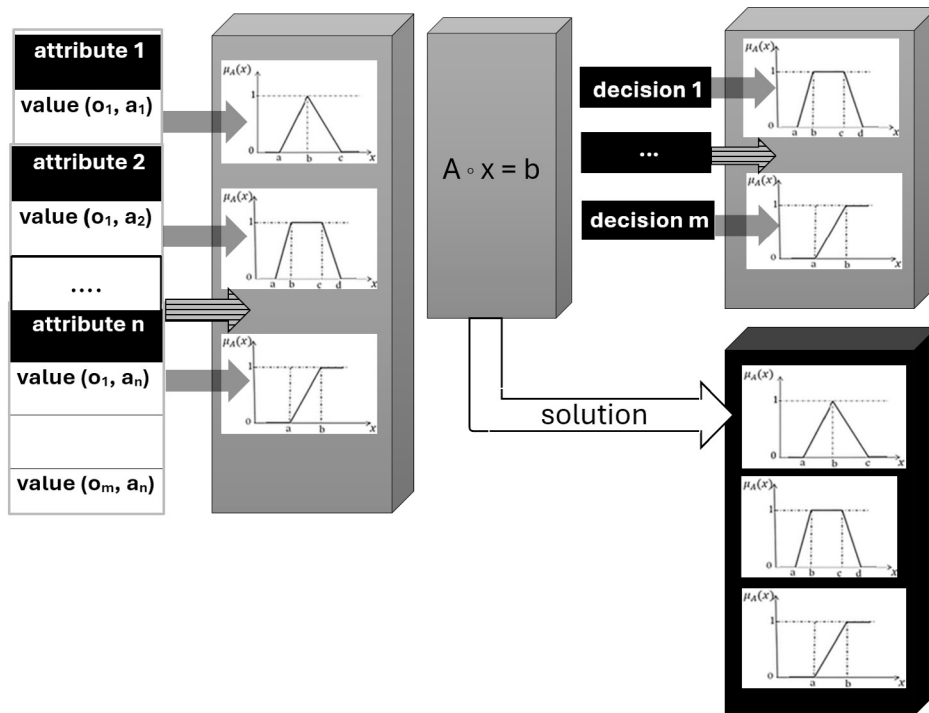


Fig. 2. Inverse problem resolution.

data. It takes place in the following steps:

1. Translating linguistic values (both attributes and decisions) into values in the range $[0, 1]$, i.e. establishing and using the membership function;
2. Solving a relational equation (a system of equations of the $\max - *$ type or appropriate);
3. Translation of results into linguistic values.

Example 1.1. For example, the data table may contain attributes as in Table 1: Headache, Muscle-pain, Temperature and Flu Decision. However, the data are more detailed in Table 2.

Table 2. An example of a decision table with some linguistic values.

Patient	Headache	Muscle pain	Temperature	Flu
p1	little	little	39°	possible
p2	very slight	large	40°	highly likely
p3	slight	no	37°	impossible
p4	no	large	37°	impossible
p5	large	very large	37.5°	unlikely
p6	no	very large	40°	yes

This can be translated into a table of values (depending on the set membership functions) for Table 3.

Table 3. An example of a decision table with values from $[0, 1]$.

Patient	Headache	Muscle pain	Temperature	Flu
p1	0.4	0.4	0.8	0.7
p2	0.1	0.9	1	0.6
p3	0.2	0	0	0
p4	0	0.9	0	0
p5	1	1	0.2	0.2
p6	0	1	1	1

The search for a relationship, therefore, consists of solving the system $A \circ x = b$, where

$$A = \begin{bmatrix} 0.4 & 0.4 & 0.8 \\ 0.1 & 0.9 & 1 \\ 0.2 & 0 & 0 \\ 0 & 0.9 & 0 \\ 1 & 1 & 0.2 \\ 0 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 0.7 \\ 0.6 \\ 0 \\ 0 \\ 0.2 \\ 1 \end{bmatrix}.$$

The determination of the vector x serves to find the relationships between the input data and the output data.

2. The process of finding the best relationships between input and output based on systems of equations

In this section, we recall the basics of relational systems of equations, present a new type of relations $\max - \{*_1, \dots, *_n\}$, $n \in \mathbb{N}$ and indicate the applications of these systems.

2.1. Mathematical foundations

Let $m, n \in \mathbb{N}$, $M = \{1, \dots, m\}$, $N = \{1, \dots, n\}$. Vectors $x, y \in [0, 1]^n$ and matrices $A, B \in [0, 1]^{m \times n}$ are ordered by

$$(x \leq y) \Leftrightarrow (x_j \leq y_j), (A \leq B) \Leftrightarrow (a_{ij} \leq b_{ij}), i \in M, j \in N. \quad (2)$$

The symbol $\vee, \wedge$ denotes the maximum and minimum operations (respectively).

Since we write fuzzy relational equations in matrix form, the next part of this work uses algebraic notation. So, for the given matrix $A \in [0, 1]^{m \times n}$, the vector $b \in [0, 1]^m$, and the searched vector $x \in [0, 1]^n$, in the classical form, we focus on solving systems of equations of the form:

$$\bigvee_{k \in N} a_{ik} * x_k = b_i, \quad i \in \{1, \dots, m\}. \quad (3)$$

Many authors have researched the existence of solutions and the form of a set of all solutions to a fuzzy system of equations, where the operation $*$: $[0, 1]^2 \rightarrow [0, 1]$ is a certain increasing operation. Particular attention is paid to the case when $*$ = min. The literature provides a formula for calculating the greatest solution for that type of system [22], and many methods for determining minimum solutions have been proposed. The properties of these solutions depend on the operation $*$. To summarize, this problem was analyzed for $*$ = T_P [2] and more generally for continuous, triangular norms [18], [17], [23], continuous, strict, monotonic operations $*$ [4], [5], [14] or for averages [6]. The

operation $*$ (also because of the applications) had different forms. So, systems of equations with triangular norms [23], [26], [20] pseudo-t-norms [7] and many other operations have been considered. The paper [12] is an interesting report collecting information on fuzzy relational equations up to 2010. This article shows that we can use different operations simultaneously in one system.

In this work, we want to extend the considered problem to a system of the form: $A \bullet x = b$, where $\bullet = \max - \{*_1, \dots, *_n\}$. The operation $\bullet$ is called the matrices multi-operations product $A \in [0, 1]^{m \times n}$ and $E \in [0, 1]^{n \times p}$, where

$$(A \bullet E)_{ij} = \bigvee_{k \in N} (a_{ik} *_k e_{kj}). \quad (4)$$

Lemma 2.1. *If all the operations $*_1, \dots, *_n : [0, 1]^2 \rightarrow [0, 1]$, then the matrices multi-operations product $\bullet$ is increasing, too:*

$$(A \leq C) \Rightarrow (D \bullet A \leq D \bullet C), \quad (A \leq C) \Rightarrow (A \bullet E \leq C \bullet E), \quad (5)$$

where $A, C \in [0, 1]^{m \times n}$ and $D \in [0, 1]^{r \times m}$, $E \in [0, 1]^{n \times p}$.

Proof. Because $(A \leq C)$ so, for each element we have $a_{ij} \leq c_{ij}$, $i \in M$, $j \in N$. Since all operations are increasing, we obtain $d_{li} *_i a_{ij} \leq d_{li} *_i c_{ij}$ and $a_{ij} *_j e_{jt} \leq c_{ij} *_j e_{jt}$. Based on the maximum properties we have

$$\bigvee_{i \in M} d_{li} *_i a_{ij} \leq \bigvee_{i \in M} d_{li} *_i c_{ij}, \quad \bigvee_{j \in N} a_{ij} *_j e_{jt} \leq \bigvee_{j \in N} c_{ij} *_j e_{jt}$$

for $i \in M$, $j \in N$, $l \in \{1, \dots, r\}$, $t \in \{1, \dots, p\}$. So, the matrices multi-operations product $\bullet$ (4) is increasing. $\square$

Because for each column of the A matrix (describing a certain property of objects), we can use different $*_k$ operators. It should be emphasized here that such an approach results from the fact that individual features of objects may have different effects on the output results (decisions). So the classic fuzzy equation (3) is replaced by the following:

$$\bigvee_{k \in N} a_{ik} *_k x_k = b_i, \quad i \in \{1, \dots, m\}, \quad (6)$$

where $A \in [0, 1]^{m \times n}$, $b \in [0, 1]^m$, and the searched vector $x \in [0, 1]^n$. The set of all vectors that are solutions of $A \bullet x = b$ is denoted by $S(A, b, \bullet)$. We define the appropriate system of inequalities similarly:

$$\bigvee_{k \in N} a_{ik} *_k x_k \leq b_i, \quad i \in \{1, \dots, m\}, \quad (7)$$

and the set of all vectors that are solutions of $A \bullet x \leq b$ is denoted by $S_{\leq}(A, b, \bullet)$.

2.2. Induced implication and its role in solving equations

In order to determine the greatest solution of equation $A \bullet x = b$, we need to use the implication induced by binary operations. In fuzzy sets, we consider operations in $[0, 1]$, so the subject of our considerations are the operations

$*_k : [0, 1]^2 \rightarrow [0, 1]$ and their induced implications $\overset{*}{\rightarrow}_k : [0, 1]^2 \rightarrow [0, 1]$ and the dual induced implications of $\overset{*}{\leftarrow}_k : [0, 1]^2 \rightarrow [0, 1]$, $k \in N$.

The topic of fuzzy implications is well-researched and described (see [11], [1]), so this article recalls only the necessary definitions, properties, and examples.

Definition 2.2 ([11], Definition 1.1). A triangular norm is a binary operation $*$: $[0, 1]^2 \rightarrow [0, 1]$, which is commutative, associative, and increasing, with the neutral element $e = 1$.

Definition 2.3 ([3], Definition1). Let $*$: $[0, 1]^2 \rightarrow [0, 1]$. The binary operations $\overset{*}{\rightarrow} : [0, 1]^2 \rightarrow [0, 1]$ and $\overset{*}{\leftarrow} : [0, 1]^2 \rightarrow [0, 1]$ are called induced implication and dual induced implication by the operation $*$, where

$$a \overset{*}{\rightarrow} b = \max\{t \in [0, 1] : a * t \leq b\}, \quad a \overset{*}{\leftarrow} b = \min\{t \in [0, 1] : a * t \geq b\} \quad (8)$$

if they exist for $a, b \in [0, 1]$.

Lemma 2.4 ([14], Lemma 4). Let $\{t \in [0, 1] : a * t \leq b\} \neq \emptyset$ for $a, b \in [0, 1]$. If an increasing operation $*$: $[0, 1]^2 \rightarrow [0, 1]$ is left continuous on the second argument, then it induces implication (8) in $[0, 1]$.

Corollary 2.5 ([14], Corollary 1). If an increasing operation $*$: $[0, 1]^2 \rightarrow [0, 1]$ is left continuous on the second argument and $1 * 0 = 0$, then it induces implication (8) in $[0, 1]$.

Lemma 2.6 ([14], Lemma 5). Let $\{t \in [0, 1] : a * t \geq b\} \neq \emptyset$ for $a, b \in [0, 1]$. If an increasing operation $*$: $[0, 1]^2 \rightarrow [0, 1]$ is right continuous on the second argument, then it induces implication (8) in $[0, 1]$.

Lemma 2.7 (Drewniak-84, Theorem 2). . Let $*$: $[0, 1]^2 \rightarrow [0, 1]$ be an increasing operation and $a, b \in [0, 1]$. If the induced implication (8) exists, then it has the following properties:

$$a * (a \overset{*}{\rightarrow} b) \leq b \leq a \overset{*}{\leftarrow} (a * b), \quad (9)$$

$$(a * c \leq b) \Leftrightarrow (c \leq a \overset{*}{\rightarrow} b). \quad (10)$$

Let us recall the most popular operations that satisfy these properties. It should be mentioned, however, that operations other than triangular operations, including averages and pseudo-triangular norms, are also used.

Example 2.8. The triangular norms and their implications and dual implications (for $x \geq y$):

$T_M(x, y) = \min(x, y)$ (minimum),	$x \overset{T_M}{\rightarrow} y = \begin{cases} 1 & \text{if } x \leq y \\ y & \text{otherwise} \end{cases}$	$x \overset{T_M}{\leftarrow} y = y,$
$T_P(x, y) = x \cdot y$ (product),	$x \overset{T_P}{\rightarrow} y = \begin{cases} 1 & \text{if } x \leq y \\ \frac{y}{x} & \text{otherwise} \end{cases}$	$x \overset{T_P}{\leftarrow} y = \begin{cases} \frac{y}{x} & \text{if } x \neq 0 \\ 0 & \text{otherwise} \end{cases}$
$T_L(x, y) = \max(0, x + y - 1)$ (Łukasiewicz),	$x \overset{T_L}{\rightarrow} y = \min(1 - x + y, 1)$	$x \overset{T_L}{\leftarrow} y = \begin{cases} \min(1, 1 - x + y) & \text{if } y \neq 0 \\ 0 & \text{otherwise} \end{cases}$

(11)

2.3. The greatest solution

Similarly to binary operations, we also define the concept of induced implication for the $\bullet$ operation (similarly to what was done in [14]):

$$A \overset{\bullet}{\rightarrow} b = \max\{x \in [0, 1]^n : A \bullet x \leq b\}, \quad (12)$$

Let $u \in [0, 1]^n$, where

$$u_j = \bigwedge_{i \in M} a_{ij} \overset{*_j}{\rightarrow} b_i, \quad j \in N. \quad (13)$$

Definition 2.9. By the greatest solution of system $A \bullet x \leq b$ ($A \bullet x = b$) with $\max\{-\{*_1, \dots, *_n\}$ product we call the greatest element in $S_{\leq}(A, b, \bullet)$ (in $S(A, b, \bullet)$) with respect to the partial order (2) (if it exists).

Lemma 2.10. If the binary operations $*_1, \dots, *_n : [0, 1]^2 \rightarrow [0, 1]$ are increasing, left-continuous on the second argument and $1 *_k 0 = 0$ for $k \in N$, then the zero vector is the least solution of $S_{\leq}(A, b, \bullet)$.

Proof. Since all operations are increasing and maximum have the following property: $a_{ik} = \bigvee_{j \in N} a_{ij}$, for certain $k \in N$, the inequality (7) has the following form:

$$\bigvee_{j \in N} (a_{ij} *_j 0) = a_{ik} *_k 0 \leq 1 *_k 0 = 0 \leq b_i,$$

for each $i \in M$. Therefore, the zero vector is the solution of $A \bullet x \leq b$. □

Theorem 2.11. If the binary operations $*_1, \dots, *_n : [0, 1]^2 \rightarrow [0, 1]$ are increasing, left-continuous on the second argument and $1 *_k 0 = 0$ for $k \in N$, then $u = A \overset{\bullet}{\rightarrow} b$ is the greatest solution of the inequality $A \bullet x \leq b$.

Proof. Based on Lemma 2.4, we know that for each operation $*_k$, $k \in N$ exist induced implications by them. Therefore, we can compute u . Now, from the maximum property and (9), we know that

$$(A \bullet u)_j = \bigvee_{i \in M} (a_{ij} *_j \bigwedge_{l \in M} (a_{il} \overset{*_j}{\rightarrow} b_l)) \leq \bigvee_{i \in M} (a_{ij} *_j (a_{il} \overset{*_j}{\rightarrow} b_l)) \leq \bigvee_{i \in M} (b_i) = b_j.$$

So $u \in S_{\leq}(A, n, \bullet)$.

Let x be any solution of $A \bullet x \leq b$. From (10), we get

$$(a_{ij} *_j x_j \leq b_i) \Leftrightarrow (x_j \leq (a_{ij} \overset{*_j}{\rightarrow} b_i)),$$

for $i \in M$. Since this property holds for every i , then from the minimum property we have:

$$x_j \leq \bigwedge_{i \in M} (a_{ij} \overset{*_j}{\rightarrow} b_i) = u_j.$$

Therefore, u is the greatest solution of $A \bullet x \leq b$. □

Theorem 2.12. Let $S(A, b, \bullet) \neq \emptyset$. If the binary operations $*_1, \dots, *_n : [0, 1]^2 \rightarrow [0, 1]$ are increasing, left-continuous on the second argument and $1 *_k 0 = 0$ for $k \in N$, then $u = A \dot{\rightarrow} b$ is the greatest solution of the equation $A \bullet x = b$.

Proof. Since $S(A, b, \bullet) \neq \emptyset$, so exists $x \in S(A, b, \bullet)$. From Theorem 2.11 and Lemma 2.1 we get

$$b = A \bullet x \leq A \bullet u \leq b.$$

Therefore, u is the greatest solution of $A \bullet x = b$. □

Dual implications induced by binary operations in classical relational systems of equations are used to determine the minimal solutions. In further works, we will discuss the problem of the existence and determination of minimal solutions for these new systems of equations.

Example 2.13. Note that some operations can be ordered. For example, we have:

$$T_L \leq T_P \leq T_M$$

for the triangular norms from Example 2.8. Therefore, if we assume attribute one is the most important, then $*_1 = T_M$ and the least essential third attribute is $*_2 = T_P$, $*_3 = T_L$. Let us consider the following computational example:

$$A = \begin{bmatrix} 0.4 & 1 & 0.2 \\ 0.3 & 0.2 & 0.8 \\ 0.4 & 0.9 & 0.4 \\ 0.5 & 0.6 & 0.9 \end{bmatrix}, \quad b = \begin{bmatrix} 0.5 \\ 0.64 \\ 0.45 \\ 0.72 \end{bmatrix},$$

based on (13), we compute the greatest solution u of equation $A \bullet x = b$, and we illustrate the process of multi-operations multiplications

$$u = \begin{bmatrix} 1 \\ 0.5 \\ 0.82 \end{bmatrix}, \quad A \bullet u = \begin{bmatrix} 0.4 \vee 0.5 \vee 0.02 \\ 0.3 \vee 0.1 \vee 0.64 \\ 0.4 \vee 0.45 \vee 0.22 \\ 0.5 \vee 0.3 \vee 0.72 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.64 \\ 0.45 \\ 0.72 \end{bmatrix}.$$

The results should be interpreted so that the greatest relationship between the first attribute and the decision is 1, between the second attribute and the decision is 0.5, and the third one is 0.82. The entire family of solutions must be examined to fully assess the relationship between input and output data.

However, it should be emphasised that other operations (satisfying the assumptions of Theorem 2.12) can also be used, including those that are not comparable and are not triangular norms. So, let us consider again the matrix A and the vector b with Einstein's triangular norm $*_1 = T_E$, a certain non-commutative operation $*_2 = *_{n-c}$ and the

geometric mean $*_3 = *_{gm}$, where

$$T_E(x, y) = \frac{xy}{2-x-y+xy}, \quad x \xrightarrow{*_{TE}} y = \begin{cases} 1, & x \leq y \\ \frac{2y-xy}{x+y-xy}, & \text{otherwise,} \end{cases}$$

$$x *_{n-c} y = x \cdot \max(x, y), \quad x \xrightarrow{*_{n-c}} y = \begin{cases} 1, & x^2 \leq y \\ \frac{y}{x}, & \text{otherwise,} \end{cases}$$

$$x *_{gm} y = \sqrt{xy}, \quad x \xrightarrow{*_{gm}} y = \begin{cases} 1, & x \leq y^2 \\ \frac{y^2}{x}, & \text{otherwise,} \end{cases}$$

From formula (13) we have

$$\begin{aligned} u_1 &= \min(1, 1, 1, 1) = 1 \\ u_2 &= \min(0.5, 1, 0.5, 1) = 0.5 \\ u_3 &= \min(1, 0.512, 0.50625, 0.576) = 0.50625 \end{aligned}, \quad u = \begin{bmatrix} 1 \\ 0.5 \\ 0.50625 \end{bmatrix}$$

Now, we check if u is the solution to this system of equations:

$$A \bullet u = \begin{bmatrix} T_E(0.4, 1) \vee 1 \cdot \min(1, 0.5) \vee \sqrt{0.2 \cdot 0.50625} \\ T_E(0.3, 1) \vee 0.2 \cdot \min(0.2, 0.5) \vee \sqrt{0.8 \cdot 0.50625} \\ T_E(0.4, 1) \vee 0.9 \cdot \min(0.9, 0.5) \vee \sqrt{0.4 \cdot 0.50625} \\ T_E(0.5, 1) \vee 0.6 \cdot \min(0.6, 0.5) \vee \sqrt{0.9 \cdot 0.50625} \end{bmatrix} = \begin{bmatrix} 0.4 \vee 0.5 \vee 0.45 \sqrt{0.5} \\ 0.3 \vee 0.04 \vee 0.45 \sqrt{2} \\ 0.4 \vee 0.45 \vee 0.45 \\ 0.5 \vee 0.3 \vee 0.675 \end{bmatrix} \leq \begin{bmatrix} 0.5 \\ 0.6364 \\ 0.45 \\ 0.675 \end{bmatrix} \leq \begin{bmatrix} 0.5 \\ 0.64 \\ 0.45 \\ 0.72 \end{bmatrix}.$$

Therefore, the solution set of such a system is empty. The vector u is the greatest solution of the inequality $A \bullet x \leq b$.

3. Summary

This article presents a new type of equations $A \bullet x = b$, for which product $\bullet$ has the following form $\max\{-*_1, \dots, *_n\}$, $n \in \mathbb{N}$, which have their bases in relational systems of equations. Relational equations have numerous applications, including in fuzzy reasoning. The presented results make it possible to find the greatest relationship that may exist between the input data and the output data. This model considers that there may be different relationships between different attributes and the decision. The next research stage on the presented issue will focus on determining the entire family of solutions of the system $\max\{-*_1, \dots, *_n\}$, $n \in \mathbb{N}$. It should be emphasized that examining the relationship between input and output attributes may be performed as a pre-processing stage. Determining sets of attributes that have the most decisive impact on the decision process and possible reduction of those with the least negligible impact may contribute to more effective information processing and obtaining rules from data.

Acknowledgements

This work was supported by the grant: *Balanced fuzzy sets in the development of real-world models*, University of Information Technology and Management, Poland.

References

- [1] Baczyński, M., Jayaram, B. (2008). *Fuzzy Implications*, Springer, Berlin.
- [2] Bourke, M. M., Fisher, D. G. (1998). Solution algorithms for fuzzy relational equations with max-product composition. *Fuzzy sets and Systems*, 94(1), 61-69.
- [3] Drewniak, J. (1984). Fuzzy relation equations and inequalities. *Fuzzy Sets and Systems*, 14(3), 237-247.
- [4] Drewniak, J., Matusiewicz, Z. (2010). Properties of fuzzy relation equations. *Soft Computing*, 14(10), 1037-1041.
- [5] Drewniak, J., Matusiewicz, Z. (2012). Fuzzy equations $\max *$ with conditionally cancellative operations. *Information Sciences*, 206, 18-29.
- [6] Ghodousian, A., Khorram, E. (2006). Solving a linear programming problem with the convex combination of the $\max$ - $\min$ and the $\max$ -average fuzzy relation equations. *Applied Mathematics and computation*, 180(1), 411-418.
- [7] Han, S. C., Li, H. X., Wang, J. Y. (2006). Resolution of finite fuzzy relation equations based on strong pseudo-t-norms, *Applied Mathematics Letters*, 19(8), 752-757.
- [8] Jensen, R., Shen, Q. (2004). Fuzzy-rough attribute reduction with application to web categorization, *Fuzzy Sets and Systems*, 141 (3), 469–485.
- [9] Jensen, R., Shen, Q. (2008). New approaches to fuzzy-rough feature selection. *IEEE Transactions on fuzzy systems*, 17(4), 824-838.
- [10] Kerre, E. E. (1990). A walk through fuzzy relations and their application to information retrieval, medical diagnosis and expert systems,” [1990] *Proceedings. First International Symposium on Uncertainty Modeling and Analysis*, College Park, MD, USA, 1990, pp. 84-89, doi: 10.1109/ISUMA.1990.151228.
- [11] Klement, E.P., Mesiar, R., Pap, E. (2000). *Triangular Norms*, Kluwer Academic Publishers, Dordrecht.
- [12] Li, P., Fang, S. C. (2009). A survey on fuzzy relational equations, part I: classification and solvability. *Fuzzy Optimization and Decision Making* 8, 179-229.
- [13] Lin, T. Y., Cercone, N., Skowron, A., Polkowski, L. (1997). Synthesis of decision systems from data tables. *Rough sets and data mining: Analysis of imprecise data*, 259-299.
- [14] Matusiewicz, Z., Drewniak, J. (2013). Increasing continuous operations in fuzzy $\max *$ equations and inequalities. *Fuzzy Sets and Systems*, 232, 120-133.
- [15] Matusiewicz, Z. (2023). Methods for solving balanced fuzzy systems of equation. *Procedia Computer Science*, 225, 3621-3630.
- [16] Matusiewicz, Z., Homenda, W. (2022). Balanced Relations Equations, 2022 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE), Padua, Italy, 2022, pp. 1-8, doi: 10.1109/FUZZ-IEEE55066.2022.9882798.
- [17] Miyakoshi, M., Shimbo, M. (1985). Solutions of composite fuzzy relational equations with triangular norms. *Fuzzy Sets and Systems*, 16(1), 53-63.
- [18] Pedrycz, W. (1982). Fuzzy relational equations with triangular norms and their resolutions. *Busefal*, 11, 24-32.
- [19] Peeva, K., Kyosev, Y. (2004). *Fuzzy relational calculus: theory, applications and software (with CD-ROM)* (Vol. 22). World Scientific.
- [20] Peeva, K. (2013). Resolution of fuzzy relational equations—method, algorithm and software with applications. *Information Sciences*, 234, 44-63.
- [21] Sanchez, E. (1979). Inverses of fuzzy relations. Application to possibility distributions and medical diagnosis, *Fuzzy Sets and Systems* 2 (1), 75-86.
- [22] Sanchez, E. (1976). Resolution of composite fuzzy relation equations. *Information and control*, 30(1), 38-48.
- [23] Shieh, B. S. (2007). Solutions of fuzzy relation equations based on continuous t-norms. *Information Sciences*, 177(19), 4208-4215.
- [24] Yang, X., Chen, H., Li, T., Luo, Ch. (2022). A noise-aware fuzzy rough set approach for feature selection, *Knowledge-Based Systems* 250, 109092.
- [25] Yuan, Z., Chen, H., Xie, P., Zhang, P., Liu, J., Li, T. (2021). Attribute reduction methods in fuzzy rough set theory: An overview, comparative experiments, and new directions, *Applied Soft Computing* 107, 107353.
- [26] Wu, Y. K., Guu, S. M. (2008). An efficient procedure for solving a fuzzy relational equation with $\max$ -archimedean t-norm composition, *IEEE Transactions on Fuzzy Systems*, 16(1), 73-84.